Journal of Molecular Graphics and Modelling 144 (2026) 109303

Contents lists available at ScienceDirect

Journal of Molecular Graphics and Modelling

journal homepage: www.elsevier.com/locate/jmgm

DeepHybridCPI: A hybrid deep learning framework for compound–protein interaction prediction

Areen Rasool, Jamshaid Ul Rahman *, Qasim Ali

Abdus Salam School of Mathematical Sciences GC University, Lahore, 54600, Pakistan

ARTICLE INFO

Keywords:
Compound-protein interaction
Multiscale graph neural network
Convolutional neural networks
Long short-term memory
Drug discovery

ABSTRACT

In bioinformatics, deep learning-based methods for Compound-Protein Interaction (CPI) prediction play a vital role in virtual screening, drug discovery, and drug repositioning. Recent improvements in computational methods have shown great possibility to save costs of experiment and speed up target identification. Nevertheless, the current CPI forecasting methods remain severely limited. Many rely on shallow Graph Neural Networks (GNNs) that struggle to capture the global structural context of compounds, while conventional Convolutional Neural Networks (CNNs) focus primarily on local sequence motifs and fail to model long-range dependencies in proteins. Even though a number of recent architectures strive to solve these problems by adding complexity to models, or by adding complex modules, these additions often cause significant computational overhead. To overcome these challenges, we propose DeepHybridCPI, a hybrid deep learning framework designed for accurate and efficient CPI prediction. Our hybrid model integrates a multiscale, densely connected GNN to extract compound features capturing both local substructures and global molecular topology, and employs CNNs with Long Short-Term Memory (LSTM) networks to model both local motifs and extended dependencies in protein sequences. The learned compound and protein representations are fused into a unified latent space to enable effective interaction modeling. Experimental evaluations on benchmark Human and *C. elegans* datasets demonstrate that DeepHybridCPI consistently outperforms existing state-of-the-art baseline methods in terms of AUC, Precision, and Recall. These findings highlight the importance of combining multiscale compound representations with hybrid sequence encoders within a single unified framework, providing a promising avenue for accelerating computational drug discovery. We release our source code and dataset at: https://github.com/jamshaidwarraich/DeepHybridCPI.

1. Introduction

Identifying CPIs plays a crucial role in areas such as drug repurposing and the assessment of potential side effects [1,2]. Although laboratory-based assays can give good values of the compound-protein binding affinity, they are frequently constrained by their expense and time requirements [3]. To address these challenges, a wide range of computational approaches for CPI prediction have been developed in recent years, aiming to reduce research costs, improve the efficiency of drug discovery, and speed up the overall process of drug development [4].

Deep learning has seen massive application in various fields of research, with particularly prominent applications in the field of drug discovery [5,6], where it assists with important tasks such as drug-target affinity estimation [7,8], protein-protein interaction analysis [9], and molecular property prediction [10]. In recent years, deep learning has been widely applied to CPI prediction [11–13]. Early approaches typically relied on predefined molecular fingerprints and protein descriptors generated through fixed rules. However, these handcrafted features often overlooked structural details, which could introduce bias into the prediction process. As the field of deep learning in drug discovery has progressed, numerous end-to-end models have been proposed to predict CPI [14–19]. These frameworks are learned directly on feature representations of the raw input data and they do not rely on the manually developed descriptors.

Recent studies highlight the increasing importance of hybrid architectures and attention mechanisms for compound property prediction. One such example is the DFT_ANPD [20], which combines information of the structure of compounds, as well as semantic representations obtained with large language models through a two-way attention

* Corresponding author.
E-mail address: jamshaid@sms.edu.pk (J.U. Rahman).

https://doi.org/10.1016/j.jmgm.2026.109303
Received 28 September 2025; Received in revised form 9 December 2025; Accepted 17 January 2026
Available online 19 January 2026

mechanism, prioritizing chemically pertinent information to reliably predict activity. Likewise, computational drug design systematic reviews [21] have highlighted effective compound representations, hybrid deep learning architectures, and stringent performance evaluation are required to improve the predictive strength and generality across pipelines of drug discovery. These observations are inspiring the creation of hybrid models which are able to reflect both the characteristics of compounds and proteins as well as to be interpretable and computationally efficient.

For instance, DeepDTA [17] leverages CNNs to extract latent feature representations from SMILES strings [22] and protein sequences. Graph-CNN [18] concentrates on the protein pockets by modeling drug-binding sites as graphs, but uses two distinct graph convolutional neural networks (GCNNs) to extract structural information about each protein pocket graph and each 2D compound graph. Similarly, GraphDTA [23] combines GNNs for molecular feature extraction with CNNs for protein sequence modeling; their outputs are fused through a dense neural network to predict drug-target affinities. End-to-end frameworks that integrate graph-based representations and pre-trained language models have shown promising results in modeling complex molecular and protein features [24,25]. In addition, a number of end-to-end methods enhance interpretability by incorporating attention mechanisms [26], a technique broadly used in deep learning to highlight important local features [14,27–29]. For example, CPI-GNN emphasizes critical protein subsequences relevant to drug binding [14]. TransformerCPI applies an attention module to align compound features with different proteins, providing insights into whether a compound interacts with the extracellular or transmembrane regions of GPCR proteins [27]. CoaDTI introduces a co-attention strategy that simultaneously captures key drug atoms and protein residues involved in binding [28]. CmhAttCPI [30] employs GCNs for molecular graphs and CNNs for protein sequences, integrating them through cross multi-head attention followed by an FCNN. CPI-GGS [31] combines GATv2 and GCN for drug representation with CNN and BiGRU (with n-grams) for protein modeling, integrating features via self-attention and a fully connected layer. TriCvT-DTI [32] incorporates bidirectional multi-head attention to enable interactive feature learning between drugs and targets, thereby modeling complex relationships more effectively. Transformer based and multimodal based architectures have also been used to learn complex interactions between compounds and proteins with better performance than more traditional approaches [33,34].

Despite the fact that recent end-to-end frameworks have achieved promising performance in CPI prediction, their interpretability is still limited as the local important features that have a substantial contribution to CPIs are not sufficiently analyzed. Moreover, several major challenges remain underexplored. First, most existing models employ shallow GNNs to extract compound features; however, GNNs with only a few layers are inadequate for capturing the global structural information of compounds, while simply stacking more layers often leads to issues such as over-smoothing and training instability. Second, protein sequence features are often extracted using CNNs; yet CNNs are inherently limited to learning local patterns and fail to capture long-range sequential dependencies, which are essential for accurately modeling protein characteristics. Third, the interpretability of graph-based CPI models largely depends on the attention mechanism. While attention enhances visual explanations, it also significantly raises computational overhead. Furthermore, the graph attention mechanism typically employs masked attention that only considers the local neighborhood of a vertex [35,36], thereby failing to capture the global interactions among atoms within a molecule.

These drawbacks limit accurate and effective prediction of CPIs which represents the need to have a framework that at the same time provides both global structural knowledge of compounds and long-range dependencies in protein sequences without becoming computationally infeasible. To address these challenges, we propose DeepHybridCPI, a novel hybrid deep learning framework that integrates a multiscale graph neural network (MGNN) to capture simultaneously global and local compound structures, with a CNN-LSTM protein encoder which models long-distance relationships within protein sequences. In contrast to the earlier methods, DeepHybridCPI effectively extracts both comprehensive compound structures and long-range protein information within a unified framework, and also maintains critical feature details using a sine linear unit (SLU) activation function. This combination not only improves predictive accuracy but also enhances the interpretability and robustness of the learned representations.

Key contributions of this work include.

1. MGNN for molecular graphs encoding hierarchical structural features at both atom-level and global topology.
2. Protein sequence CNN-LSTM network that combines both local motifs and long-range dependencies.
3. Autonomous incorporation of compound and protein embeddings into a latent space that can be effectively used to predict CPI.
4. Extensive experimental validation on benchmark datasets (Human and *C. elegans*) demonstrating superior performance over state-of-the-art methods.

DeepHybridCPI fills some of the gaps contingent on currently available approaches and offers a comprehensive, efficient, and interpretable predictive framework of CPI that can make drug discovery pipelines faster.

2. Materials and methods

2.1. Datasets

An accurate and robust predictor requires a well-curated and robust dataset [37–39]. In machine learning-based CPI prediction, training datasets typically consist of two categories: positive samples, which are experimentally confirmed CPIs, and negative samples, which are experimentally validated non-interacting compound-protein pairs. Many earlier studies commonly generated negative samples by randomly shuffling the true CPI pairs [38]. However, such random sampling can inadvertently include unknown positive interactions, leading to variations in model performance across datasets and potentially reducing predictive reliability. Hence, a key step in constructing the dataset is to ensure selecting authentic and trustworthy negative samples.

In this study, two of the most popular CPI datasets, the Human and *C. elegans* originally curated by Liu et al. [38] are chosen as benchmark datasets based on their broad extent and their established usefulness in drug discovery studies. The datasets include negative samples that have been carefully verified through statistical analyses to ensure that compounds and proteins in these pairs do not interact, resulting in highly trustworthy non-interacting pairs. The complete procedure for selecting these negative samples is described in Ref. [14]. Each data set has three primary components: (i) drug sequences in a textual format, (ii) protein target sequences also in text format, and (iii) a binary label, indicating the existence or non-existence of a CPI. Table I shows summary statistics of the number of compounds and proteins per dataset. Both datasets maintain a balanced distribution, containing approximately equal numbers of positive (binding) and negative (non-binding) samples, with

Table 1
Summary of the distribution for the Human and *C. elegans* benchmark datasets.

Datasets	Human	*C. elegans*
Total compounds	2726	1767
Total proteins	2001	1876
Total samples	6728	7786
Positive interactions	3364	3893
Negative interactions	3364	3893

a ratio close to 1:1.

2.2. Proposed model

Fig. 1 illustrates the framework of the proposed DeepHybridCPI model, which consists of three main modules: (i) the data preprocessing module, (ii) the feature extraction module, and (iii) the prediction module. First, the data preprocessing component converts compound and protein information into an appropriate form to process it. Next, the feature extraction module generates informative feature representations from both compounds and proteins. Finally, the prediction module combines the compound and protein features to predict the interaction of the given compound-protein pair.

A detailed overview is provided of the key components that contribute to the enhanced performance of the proposed model, namely: compound feature learning, protein feature learning, and the CPI prediction network.

2.2.1. MGNN for compound feature extraction

The molecular input for the network is encoded employing the Simplified Molecular Input Line Entry System (SMILES), which utilizes concise ASCII strings to portray the chemical structures of the compounds. These SMILES strings are processed with RDKit [40] to generate graph-based representations, including node attributes and adjacency matrices. This graph is characterized by atoms being nodes and chemical bonds the edges. As shown in Fig. 2, each node is represented by an 87-dimensional feature vector capturing detailed atomic properties.

Once the compounds are represented as graphs, a MGNN [41] is employed for compound encoding, enabling the capture of both local and global structural relationships in the graph. The MGNN consists of three multiscale blocks, each of which is followed by a transition layer, which combines and reduces features and transmits them to the next level.

A multiscale block has N graph convolutional layers and each layer is characterized by the following expression:

$$m_i^{(t+1)} = \sigma\left(W_1 m_i^{(t)} + W_2 \sum_{j \in N(i)} m_i^{(t)}\right). \tag{1}$$

Here W_1, $W_2 \in \mathbb{R}^{h \times d}$ are learnable weight matrices shared across all vertices, $m_i^{(t)} \in \mathbb{R}^d$ is the embedding of the i-th vertex at time step t, $N(i)$ represents the neighbors of node i, and σ comprises a node-level batch normalization [42] followed by the SLU activation function [43], defined as:

$$f(x) = \begin{cases} x, & \text{if } x \geq 0 \\ \sin(x), & \text{if } x < 0. \end{cases} \tag{2}$$

Equation (1) aggregates information from each node's neighbors, and by repeating this process over multiple iterations, the node embeddings progressively capture more global information about the compound graph. This thick connection into GNN connects all layers to all others in a feed-forward way that enables the gradients to flow directly and simplifies the training of very deep networks. The formal statement of a multiscale block is:

$$\begin{aligned} m_i^{(1)} &= \mathrm{H}\left(m_i^{(0)}, \varnothing_1\right), \\ m_i^{(2)} &= \mathrm{H}\left(m_i^{(0)} \| m_i^{(1)}, \varnothing_2\right), \\ m_i^{(3)} &= \mathrm{H}\left(m_i^{(0)} \| m_i^{(1)} \| m_i^{(2)}, \varnothing_3\right), \\ m_i^{(N)} &= \mathrm{H}\left(m_i^{(0)} \| m_i^{(1)} \| \ldots \| m_i^{(N-1)}, \varnothing_N\right). \end{aligned} \tag{3}$$

Here H is the graph convolutional layer defined by Equation (1), $\varnothing_n$ denotes the parameters (W_1, W_1) of the n-th layer, and || represents concatenation. The multiscale block generates multiscale features that characterized the structure data of a compound, both in local and wider structural environments, which is fundamental to effective prediction of interactions.

In the MGNN, multiscale blocks are joined by transition layers at

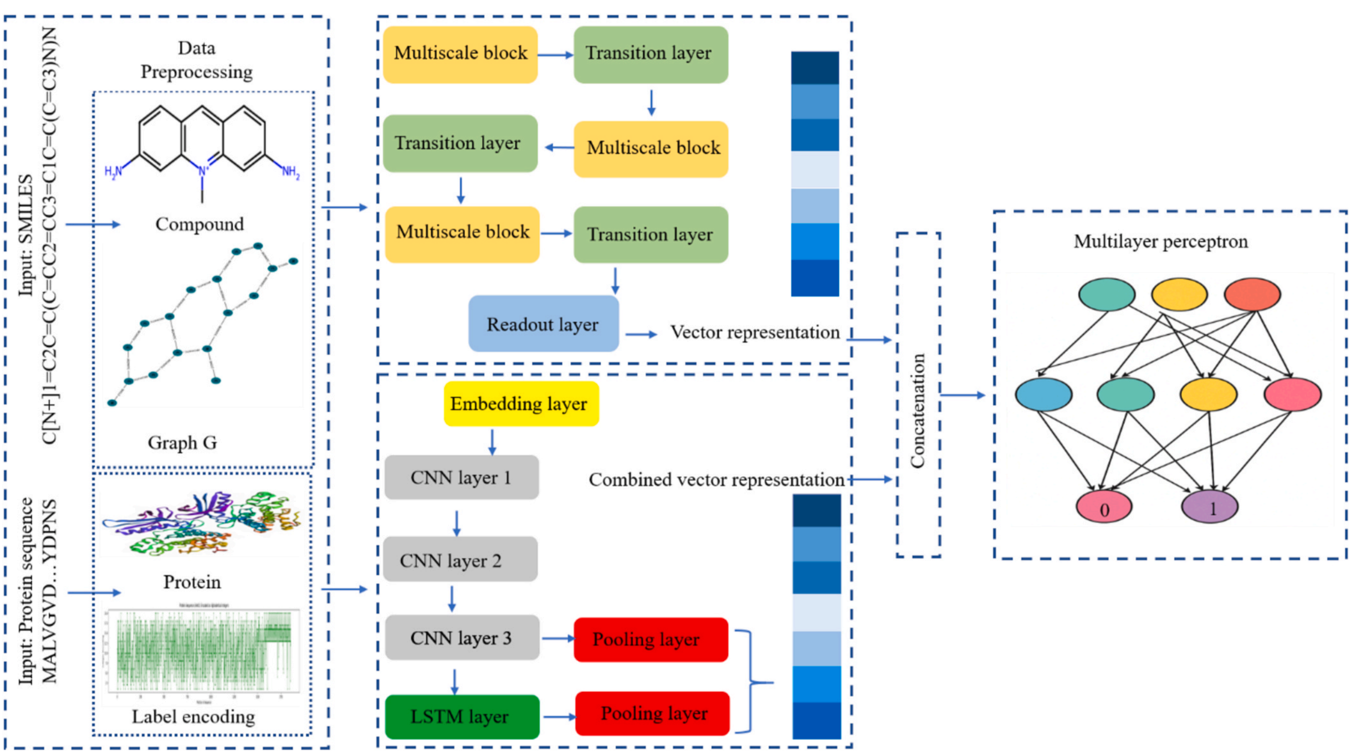

Fig. 1. Framework of DeepHybridCPI where compound graphs are encoded with MGNN and protein sequences with CNN-LSTM then concatenated and passed through fully connected layers to predict CPIs.

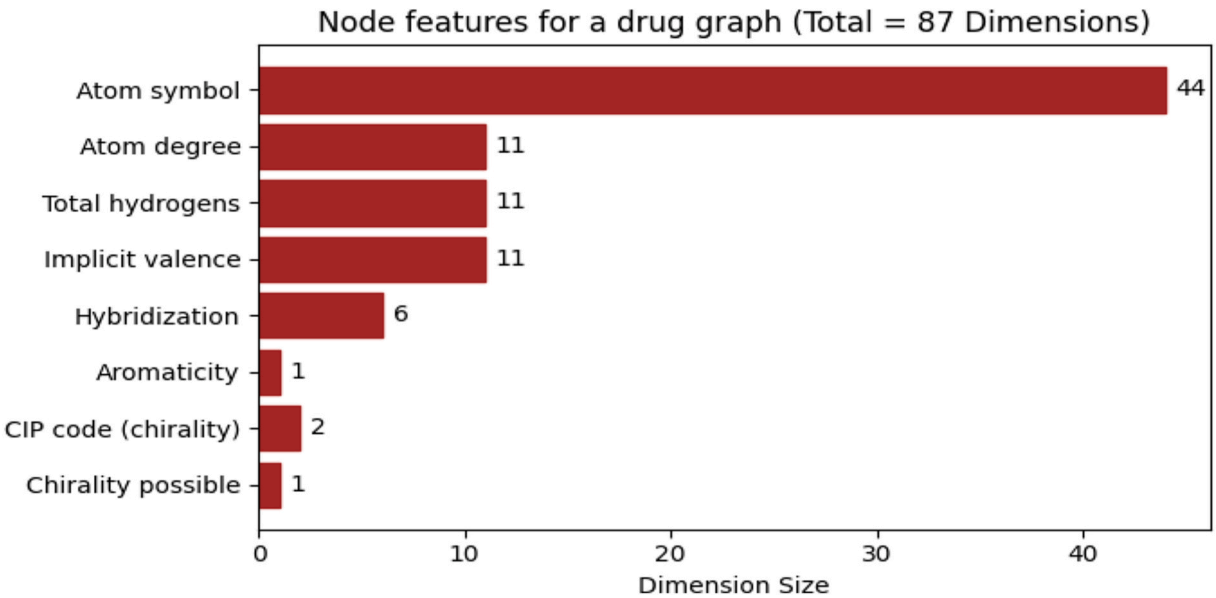

Fig. 2. Distribution of the eight atom-level features and their corresponding dimensions, used to represent each node in the compound graph leading to a one-hot vector of 87 feature dimensions.

each side. Their goal is to combine the multiscale characteristics of the former multiscale block and minimize the number of channels in the feature map.

For an input multiscale features at time step $N+1$ as $m_i^{(0)}\|m_i^{(1)}\|\ldots\| m_i^{(N)} \in \mathbb{R}^{d+(N-1)h}$; the transition layer is given by:

$$m_i^{(N+1)} = \sigma\left(\varphi_1\left(m_i^{(0)}\|m_i^{(1)}\|\ldots\| m_i^{(N)}\right) + \varphi_2 \sum_{j\in N(i)}\left(m_j^{(0)}\left\|m_j^{(1)}\right\|\ldots\| m_j^{(N)}\right)\right), \tag{4}$$

where $\varphi_1,\ \varphi_2 \in \mathbb{R}^{(M/2)\times M}$ denote learnable weight matrices that are shared across all vertices, where $M = d + (N-1)h$. By using the transition layer, we reduced the channel numbers to half of the input to save computational cost.

Lastly, a readout layer is used to aggregate the node level features into a graph level representation. The readout is the following:

$$Y_G = \frac{1}{|V|}\sum_{v\in V} m_v^{(L)}, \tag{5}$$

where $|V|$ represents the total number of nodes in the compound graph G, L is the final time step, and $Y_G \in \mathbb{R}^d$ denotes the corresponding graph-level feature vector.

2.2.2. CNN-LSTM for protein feature extraction

The raw protein sequences are fed into the model as inputs. The first step involves the transformation of these sequences via the encoding layers. A vocabulary is created that assigns a unique integer to each amino acid, allowing the protein sequence to be expressed as a sequence of integers (as demonstrated in Fig. 3).

To ensure that training is convenient, the maximum length of the sequence is set to 1200, which includes at least 80 % of the proteins in the dataset according to a previous study [42]. Shorter sequences are padded with zeros, and longer sequences are truncated. An embedding layer is then used to map each integer to a learnable 128-dimension vector [11,17], such that each amino acid is denoted by a 128-dimensional embedding vector.

The embedded protein matrix is then passed through three consecutive 1D convolutional layers. The first convolutional layer employs 32 filters with 8 kernel size, where the second and third layers employ 2–3 times more filters than the first. After the first and second convolutional layers, a dropout layer with a probability of 0.2 is incorporated to help prevent overfitting, followed by the SLU activation function defined in Equation (2). These convolutional layers successively extract local patterns in the sequence as well as enlarging the representational capacity of the model.

Next, a 1D global max pooling operation is applied to the CNN output, $X \in \mathbb{R}^{B\times C\times L}$, where B is the batch size, C is the number of channels (feature maps), and L is the sequence length after convolution. The pooling operation is as follows:

$$y_{b,c} = \max_{1\le t\le L} X_{b,c,t} \tag{6}$$

This chooses the most salient activation per feature map in each position, producing $Y^{CNN} \in \mathbb{R}^{B\times C}$.

Since CNNs primarily learn local features, an LSTM layer is applied to capture long-range sequential dependencies [44]. The CNN block output is reshaped and fed inside the LSTM which encodes both short and long dependencies in the sequence. The LSTM output is $\in \mathbb{R}^{B\times T\times H_{dim}}$, where T is the processed sequence length and H_{dim} is the dimensionality of the LSTM's hidden state (set equal to the number of filters in the final convolutional layer).

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Amino Acid	A	C	B	E	D	G	F	I	H	K	M	L	O	N	Q	P	S	R	U	T	W	V	Y	X	Z
Value	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Fig. 3. Mapping of amino acids to their unique integer identifiers used in the sequence-based protein encoder for CPI prediction.

Afterwards, a global max pooling is applied in the time-axis of the LSTM output:

$$y_{b,k}^{(LSTM)} = \max_{1\le t\le T} H_{b,t,k} \tag{7}$$

This produces $Y^{LSTM} \in \mathbb{R}^{B\times H_{dim}}$.

Lastly, along the feature dimension, the pooled CNN features Y^{CNN} and pooled LSTM features Y^{LSTM} are concatenated:

$$F_{concat} = Y^{CNN} \| Y^{LSTM} \in \mathbb{R}^{B\times 2C} \tag{8}$$

where $C = 32 \times 3$ is the number of filters in the last CNN layer. These concatenated features are fed to a fully connected layer of 96 neurons, before being forwarded to the final prediction layer.

2.2.3. DeepHybridCPI network

To fully capture the characteristics of both compounds and proteins, their respective vector representations are concatenated to form a joint representation, which is then fed into the prediction module to determine whether the given compound-protein pair will interact. Specifically, the MLP consists of four linear transformations layers of dimensions 192 → 1024, 1024 → 1024, 1024 → 256, and 2256; → 2, respectively. Every linear transformation layer is succeeded by a SLU activation and a dropout operation with a rate of 0.2.

This simple feature fusion strategy is computationally efficient and straightforward to implement compared to attention-based or transformer-based alternatives that offer better interpretability but require more data and computational resources. This method does not formulate all the intricate interactions among features, but is a sensible trade off, in that it provides a strong predictive accuracy at lower computational expense.

The cross-entropy loss function is used to train the model, which is defined as:

$$loss_{cross-entropy} = -x[class] + log\left(\sum_j \exp\,(x[j])\right) \tag{9}$$

Here, x is the vector of logit predictions of all classes on a sample, and class is the index of the ground-truth class label.

2.3. Evaluation metrics

To assess performance, this paper uses three popular evaluation measures [45,46]: Area Under Curve of Receiver Operating Characteristic (AUC), Precision, and Recall. The area under ROC (Receiver Operating Characteristic) curve [47,48] is called Area Under ROC Curve (AUC), it is a measurement of the classification power of the model. The definitions of other metrics are as follows:

$$Precision = \frac{TP}{TP + FP}, \tag{10}$$

$$Recall = \frac{TP}{TP + FN}, \tag{11}$$

TP, FN, TN and FP represent the number of true positives, false negatives, true negatives and false positives, respectively. Precision is the fraction of correctly predicted positive samples out of all samples predicted as positive, indicating the reliability of the model when labeling samples as positive. Recall is the proportion of actual positive samples that are correctly identified, reflecting the model's effectiveness in detecting positive cases.

Generally, AUC, Precision as well as Recall lie between 0.5 and 1 where a higher value indicates greater discriminative ability and better predictive performance.

3. Results and discussions

3.1. Experimental setup

Table 2 presents the complete set of hyperparameter configurations used in this study. Most parameters originate from prior empirical knowledge or systematically tuned through controlled experiments, but the learning rate is highlighted here due to its critical influence on model convergence. After extensive training and validation, the learning rate is fixed at 1e-4 to ensure stable convergence and optimal performance. The test set is not utilized during hyperparameter tuning to prevent data leakage and ensure unbiased evaluation. Figs. 4 and 5 illustrate the training and validation curves of the DeepHybridCPI model for the Human and *C. elegans* datasets, respectively. The curves correspond to the average of three random seeds, and the shaded area indicates the standard deviation, which demonstrates the stability and strength of the training process of the model. In general, the training accuracy improves gradually with epochs as the model trains on the data and the validation accuracy is minimal early and then steadily increases. The metric of loss shows the reverse tendency: the training loss steadily decreases and the validation loss tends to increase in the first few epochs and levels off or diminishes slightly since the model is overfitting. Moreover, the validation accuracy and loss show slightly fluctuations compared to training, reflecting the model's limited generalization to unseen data.

All experiments are carried out on Google Colaboratory, utilizing an NVIDIA T4 GPU, an Intel Xeon CPU, and 12 GB of RAM. The implementation is done in PyTorch with CUDA enabled, and both training and inference are executed on the same hardware setup to maintain consistency and reproducibility.

3.2. Comparison with various methods

To assess the performance of the proposed DeepHybridCPI model, its CPI prediction performance is compared with several state-of-the-art deep learning frameworks, including CPI-GNN [14], TransformerCPI [27], TrimNet [50], MDL-CPI [51], and GraphCPI [52], on the Human and *C. elegans* datasets. All experiments are repeated three times, each with a different random seed. Then, we finally reported the mean and standard deviation of results in CPI prediction. The comparative outcomes are reported in Tables 3 and 4, where the best results are highlighted in bold. For all competing methods, the results reported in their original publications are directly cited to ensure a fair comparison.

Table 3 demonstrates that DeepHybridCPI outperforms most competing approaches on the Human dataset with respect to AUC and Precision. Especially, it obtains an AUC of 0.978 and a Precision of 0.947, which are 0.4–1.9 % and 0.7–3.1 % higher than those of other models, respectively. For Recall, DeepHybridCPI achieves strong results among deep learning-based methods, although TrimNet reports a slightly higher Recall of 0.953 compared to 0.941 (a difference of about 0.01). But still it's competent and highlighting that the model successfully identifies most of the true CPI interactions. However, when

Table 2
Hyperparameter configurations of the proposed DeepHybridCPI model used in the experiments.

Hyperparameters	Setting
Epoch	100
Batch size	512
Optimizer [49]	Adam
Learning rate	0.0001
Loss function	Cross-entropy
Dropout rate	0.2
Activation function	SLU
CNN layers	3
LSTM layers	1
GCN layers	27

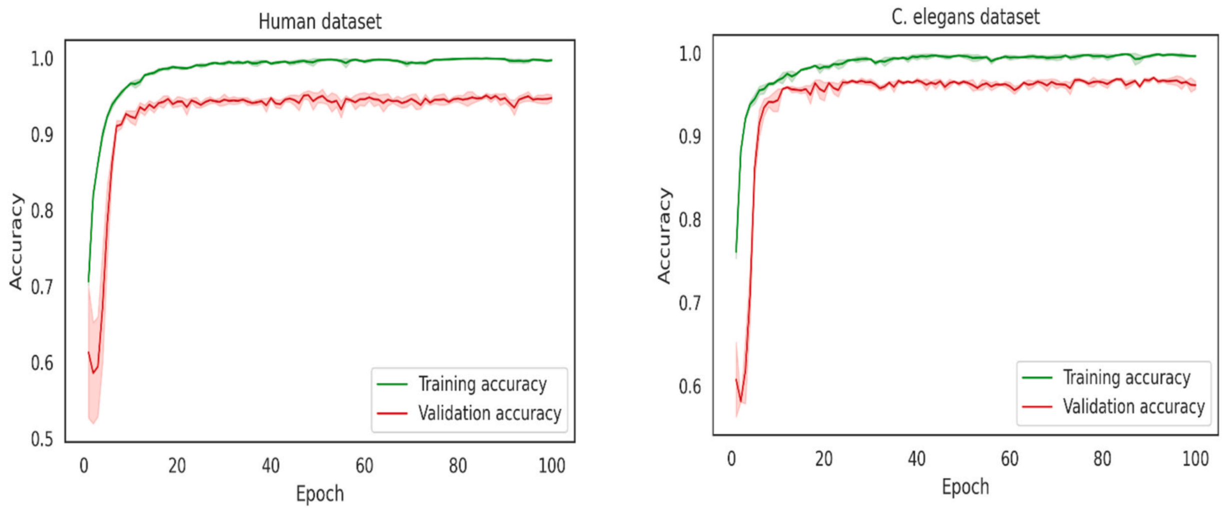

Fig. 4. Training and validation accuracy across epochs for the DeepHybridCPI model, evaluated on the Human and *C. elegans* datasets.

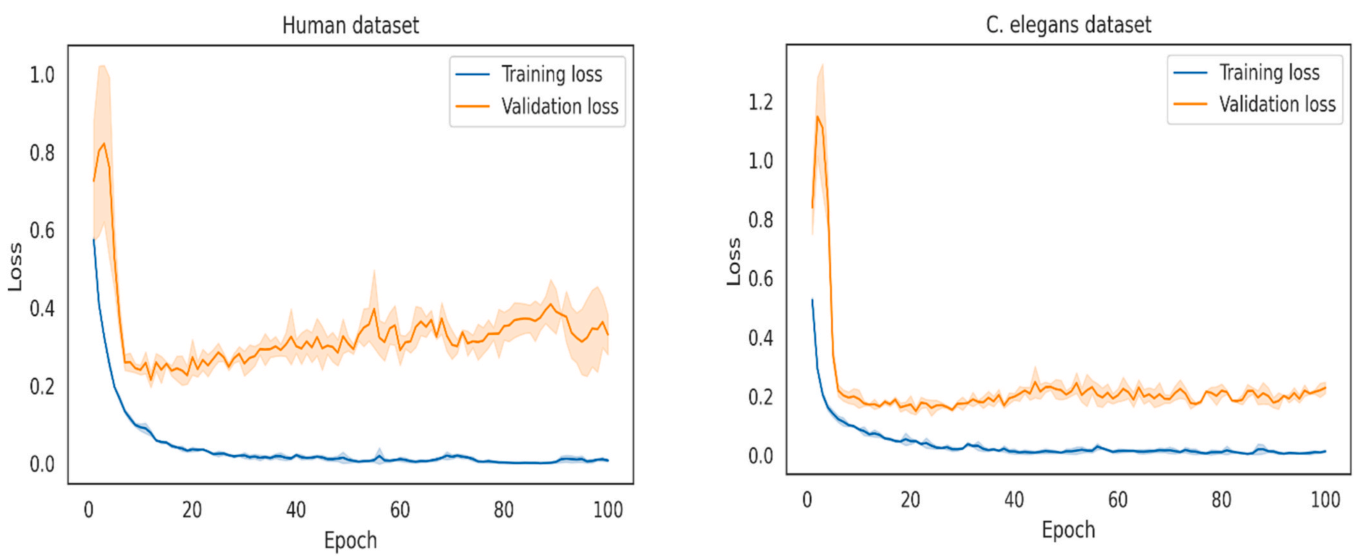

Fig. 5. Training and validation loss across epochs for the DeepHybridCPI model, evaluated on the Human and *C. elegans* datasets.

Table 3
Performance comparison of the proposed DeepHybridCPI and other baselines models on the Human dataset.

Model	AUC	Precision	Recall
CPI-GNN	0.970	0.923	0.918
TransformerCPI	0.973	0.916	0.925
TrimNet	0.974	0.918	**0.953**
MDL-CPI	0.959	0.924	0.905
GraphCPI	0.973	0.940	0.890
Our model	**0.978** ± 0.001	**0.947** ± 0.005	0.941 ± 0.0064

Table 4
Performance comparison of the proposed DeepHybridCPI and other baselines models on the *C. elegans* dataset.

Model	AUC	Precision	Recall
CPI-GNN	0.978	0.938	0.929
TransformerCPI	0.988	0.952	0.953
TrimNet	0.987	0.946	0.945
MDL-CPI	0.975	0.943	0.923
GraphCPI	0.989	0.937	0.955
Our model	**0.991** ± 0.0007	**0.967** ± 0.0051	**0.964** ± 0.0000

considering all three metrics together, DeepHybridCPI remains highly competitive and demonstrates balanced improvements across major evaluation measures. This balanced performance across AUC, Precision, and Recall positions DeepHybridCPI as a reliable and strong alternative to existing approaches.

Similarly, on the *C. elegans* dataset (Table 4), DeepHybridCPI outperforms all competing methods, achieving the highest AUC of 0.991, which is 0.2–1.6 % higher than existing approaches. It also achieves the best Precision (0.967) and Recall (0.964), demonstrating its effectiveness in extracting multimodal data features and accurately predicting true interactions. Compared to competing methods, these Precision and Recall values are higher by 0.3–3.2 % and 0.6–3.8 %, respectively, confirming the superior generalization ability of DeepHybridCPI across different datasets.

The comparisons presented above indicate that DeepHybridCPI is able to extract more informative and distinctive features for characterizing CPIs. It performs better mainly because of two reasons. On the one hand, various available procedures are based on relatively shallow GNNs, which limit the possibility to capture the global structural information of compounds. In contrast, DeepHybridCPI employs a deep

MGNN, guided by chemical intuition, to extract rich and hierarchical compound features. On the other hand, for proteins, CNN and LSTM modules are integrated, enabling the model to capture both local motifs and long-range sequence dependencies. Consequently, DeepHybridCPI is able to retain important interactive information between the compounds and proteins by concatenating these complementary representations in a merged embedding space. This integrated approach allows DeepHybridCPI to model complex interactions more effectively than methods that consider only sequence motifs or shallow graph features. Therefore, it is not surprising that this model consistently achieves superior performance compared with existing approaches.

To further assess the effectiveness of DeepHybridCPI, we applied t-SNE to visualize the learned high-dimensional representations in a two-dimensional space. Indeed, the embeddings of the interacting and non-interacting pairs of compounds and protein make distinct clusters as illustrated in Fig. 6. This clear distinction shows that the model is effective at discriminating positive and negative CPIs and is significant in distinguishing them. Furthermore, the tight grouping within each class and the clear margins between classes suggest that the learned representations are well-structured and likely to generalize effectively to unseen data.

Additionally, we also used confusion matrices to further validate the model's predictions, highlighting both correctly and incorrectly classified samples in the test results. The confusion matrix presents the number of cases correctly classified and misclassified by each of the classes (positive or negative interactions). Fig. 7 illustrates the confusion matrices of the three independent model runs, which show similar and correct patterns in prediction.

Together, these visualizations provide strong evidence that DeepHybridCPI can learn robust and biologically relevant interaction patterns.

3.3. Deformation model experiment

To examine the role of each component in the model and determine the factors responsible for optimal performance, an ablation study is conducted with four variants: (i) the full DeepHybridCPI model; (ii) DeepHybridCPI without MGNN, using only a standard GNN to extract compound features; (iii) DeepHybridCPI with ReLU activation in place of the SLU function; and (iv) DeepHybridCPI without the LSTM module, relying only on CNN and MGNN for protein feature extraction. As illustrated in Fig. 8, the complete DeepHybridCPI consistently outperforms its variants. This demonstrates the effectiveness of the hybrid learning strategy, where combining CNN-LSTM outputs with MGNN-derived features provides complementary information that significantly enhances prediction accuracy. CNN-LSTM module yields better protein embeddings than CNN, thereby capturing important sequence features better, and the multigraph embeddings generated by MGNN improve compound representation. Together, these components contribute to more accurate CPI prediction.

4. Conclusion

This study introduces DeepHybridCPI, a hybrid deep learning framework for predicting CPIs. By integrating multiscale graph-based compound representations with CNN-LSTM networks to capture protein sequence features, the model provides a richer understanding of these interactions, including whether specific compounds and proteins are likely to interact. DeepHybridCPI showed great performance on various datasets, including Human and *C. elegans* benchmarks, in terms of some metrics including AUC, Precision and Recall, highlighting its ability to play a significant role in drug discovery and development.

In addition to performance, the capability of the model to produce well separated feature embeddings indicates that it is able to capture biological signals of relevance, like structural compatibility and sequence level motifs that are related to binding. Such considerations render DeepHybridCPI applicable to practical situations, such as in virtual screening, identification of hits at an early stage, and drug-repurposing pipelines where biologically plausible interactions should play a primary role.

Several future directions exist to assess generalization and expand the model. Despite the fact that DeepHybridCPI is tested using popular Human and *C. elegans* benchmarks, subsequent studies will expand the framework to large-scale affinity datasets, including Davis, KIBA, and BindingDB, to better test the model's robustness. Its performance and flexibility could be enhanced through optimization strategies, exploring different attention mechanisms, or incorporating diverse types of biological data. Additionally, introducing the most recent approaches like self-supervised learning and multi-task learning can be useful in utilizing the large-scale biological data sets in a more efficient manner and enhancing the overall predictive performance of DeepHybridCPI.

CRediT authorship contribution statement

Areen Rasool: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Jamshaid Ul**

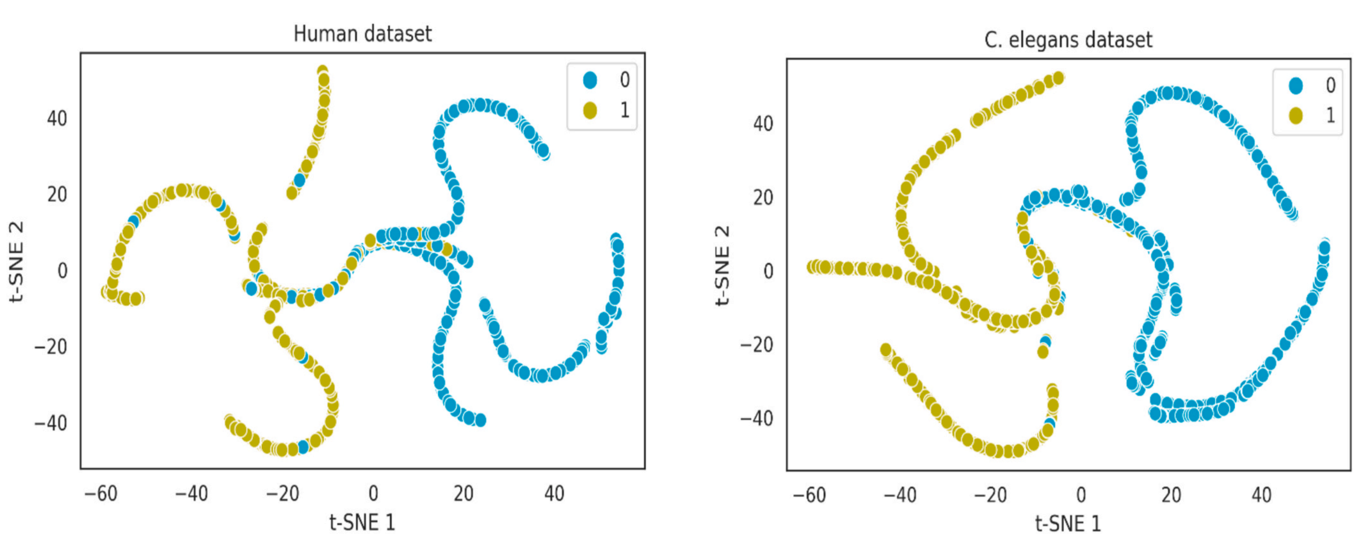

Fig. 6. t-SNE visualization shows clear separation of interacting (1) and non-interacting (0) pairs, highlighting DeepHybridCPI's discriminative features on Human and *C. elegans* test sets.

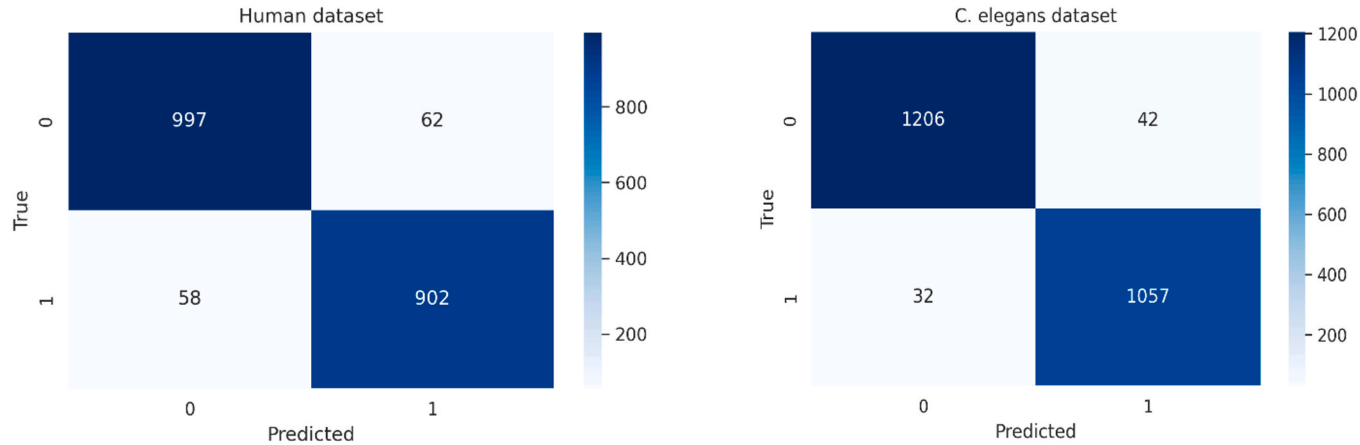

Fig. 7. DeepHybridCPI confusion matrix showing correctly and incorrectly predicted interacting (positive) and non-interacting (negative) CPIs on the Human and *C. elegans* test sets.

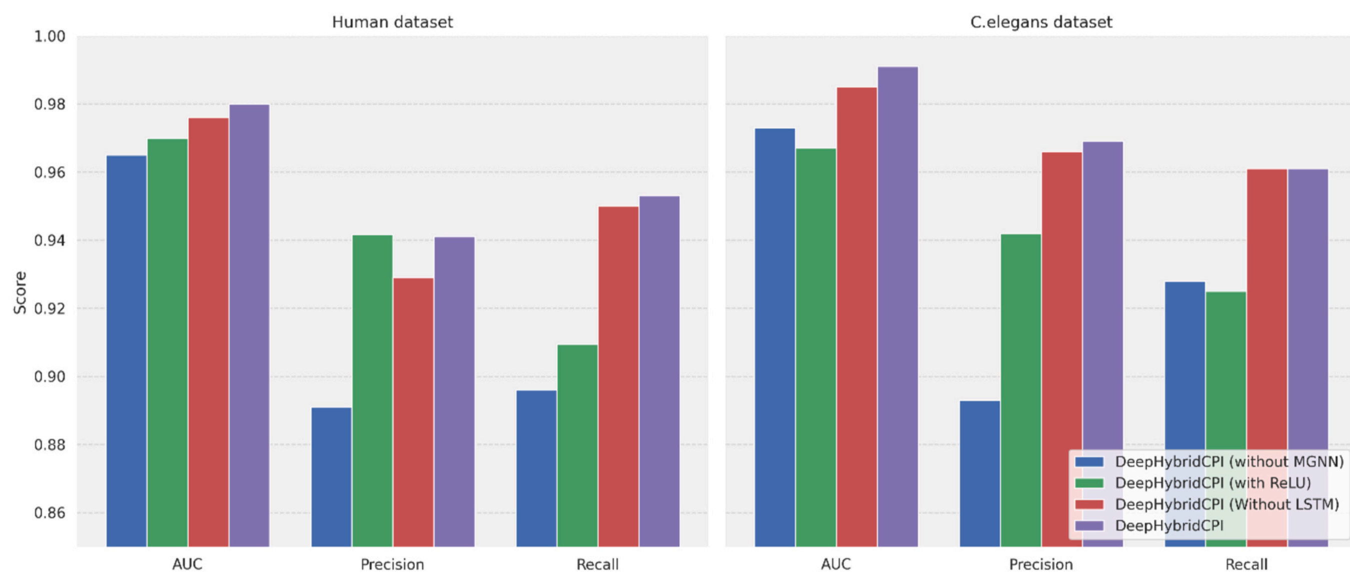

Fig. 8. Comparative results of DeepHybridCPI, DeepHybridCPI (without MGNN), DeepHybridCPI (with ReLU), and DeepHybridCPI (without LSTM) on the Human and *C. elegans* datasets.

Rahman: Writing – review & editing, Supervision, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Qasim Ali:** Writing – review & editing, Visualization, Software, Formal analysis.

Data availability statement

The datasets utilized in this study are from Liu, H., Sun, J., Guan, J., Zheng, J., & Zhou, S. (2015). Improving compound–protein interaction prediction by building up highly credible negative samples. *Bioinformatics, 31*(12), i2–i10.

Code availability

All codes and data needed to replicate the results of the present work are accessible at https://github.com/jamshaidwarraich/DeepHybridCPI.

Funding

The authors declare no funding was received for this work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets utilized in this study are from Liu, H., Sun, J., Guan, J., Zheng, J., & Zhou, S. (2015).

References

[1] S. Pushpakom, F. Iorio, P.A. Eyers, K.J. Escott, S. Hopper, A. Wells, A. Doig, T. Guilliams, J. Latimer, C. McNamee, M. Pirmohamed, Drug repurposing: progress, challenges and recommendations, Nat. Rev. Drug Discov. 18 (1) (2019) 41–58.

[2] E. Gregori-Puigjané, V. Setola, J. Hert, B.A. Crews, J.J. Irwin, E. Lounkine, L. Marnett, B.L. Roth, B.K. Shoichet, Identifying mechanism-of-action targets for drugs and probes, Proceedings of the National Academy of Sciences of the USA 109 (28) (2012) 11178–11183.

[3] S.M. Paul, D.S. Mytelka, C.T. Dunwiddie, C.C. Persinger, B.H. Munos, S. R. Lindborg, A.L. Schacht, How to improve R&D productivity: the pharmaceutical industry's grand challenge, Nat. Rev. Drug Discov. 9 (3) (2010) 203–214.

[4] M. Bagherian, E. Sabeti, K. Wang, M.A. Sartor, Z. Nikolovska-Coleska, K. Najarian, Machine learning approaches and databases for prediction of drug–target interaction: a survey paper, Briefings Bioinf. 22 (3) (2021) 247–269.
[5] J.U. Rahman, H. Ali, A. Rassol, A comprehensive analysis of optimizers in message passing neural networks for molecular property prediction task, Comput. Biol. Chem. 115 (2025) 108556.
[6] A. Rasool, J. Ul Rahman, R. Uwitije, Enhancing molecular property prediction with quantized GNN models, J. Cheminf. 17 (1) (2025). Article 81.
[7] L. Xia, L. Xu, S. Pan, D. Niu, B. Zhang, Z. Li, Drug–target binding affinity prediction using message passing neural network and self-supervised learning, BMC Genom. 24 (1) (2023) 557.
[8] S. Pan, L. Xia, L. Xu, Z. Li, SubMDTA: drug–Target affinity prediction based on substructure extraction and multi-scale features, BMC Bioinf. 24 (1) (2023) 334.
[9] X. Li, P. Han, G. Wang, W. Chen, S. Wang, T. Song, SDNN-PPI: Self-Attention with deep neural network effect on protein–protein interaction prediction, BMC Genom. 23 (1) (2022) 474.
[10] J. Ul Rahamn, M.A. Iqbal, A. Rasool, R. Uwitije, p-GIN: a graph isomorphism network based on p-laplacian operator to enhance molecular property prediction, Discov. Appl. Sci. 7 (11) (2025) 1251.
[11] I. Lee, J. Keum, H. Nam, DeepConv-DTI: prediction of drug–target interactions via deep learning with convolution on protein sequences, PLoS Comput. Biol. 15 (10) (2019) e1007129.
[12] M. Hamanaka, K. Taneishi, H. Iwata, J. Ye, J. Pei, J. Hou, Y. Okuno, CGBVS-DNN: prediction of compound–protein interactions based on deep learning, Molecular Informatics. 36 (10) (2017) 1600045.
[13] M. Wen, Z. Zhang, S. Niu, H. Sha, R. Yang, Y. Yun, H. Lu, Deep-learning-based drug–target interaction prediction, J. Proteome Res. 16 (4) (2017) 1401–1409.
[14] M. Tsubaki, K. Tomii, J. Sese, Compound–protein interaction prediction with end-to-end learning of neural networks for graphs and sequences, Bioinformatics 35 (2) (2019) 309–318.
[15] J. Wang, X. Li, H. Zhang, GNN-PT: enhanced prediction of compound–protein interactions by integrating protein transformer, arXiv Preprint (2020) arXiv: 2009.00805.
[16] T. Cai, L. Xie, S. Zhang, M. Chen, D. He, A. Badkul, Y. Liu, H.K. Namballa, M. Dorogan, W.W. Harding, End-to-end sequence-structure-function meta-learning predicts genome-wide chemical–protein interactions for dark proteins, PLoS Comput. Biol. 19 (2) (2023) e1010851.
[17] H. Oztürk, A. Özgür, E. Ozkirimli, DeepDTA: deep drug–target binding affinity prediction, Bioinformatics 34 (17) (2018) i821–i829, https://doi.org/10.1093/bioinformatics/bty593.
[18] W. Torng, R.B. Altman, Graph convolutional neural networks for predicting drug–target interactions, J. Chem. Inf. Model. 59 (10) (2019) 4131–4149.
[19] M. Karimi, D. Wu, Z. Wang, Y. Shen, DeepAffinity: interpretable deep learning of compound–protein affinity through unified recurrent and convolutional neural networks, Bioinformatics. 35 (18) (2019) 3329–3338.
[20] R. Norouzi, R. Norouzi, K. Abbasi, R. Norouzi, P. Razzaghi, DFT_ANPD: a dual-feature two-sided attention network for anticancer natural products detection, Comput. Biol. Med. 194 (2025) 110442.
[21] K. Abbasi, P. Razzaghi, A. Gharizadeh, A. Ghareyazi, A. Dehnad, H.R. Rabiee, M. R. Mofrad, Computational drug design in the AI era: a systematic review of molecular representations, generative architectures, and performance assessment, Pharmacol. Rev. (2025) 100095.
[22] D. Weininger, SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules, J. Chem. Inf. Comput. Sci. 28 (1) (1988) 31–36.
[23] T. Nguyen, H. Le, T.P. Quinn, T. Nguyen, T.D. Le, S. Venkatesh, GraphDTA: predicting drug–target binding affinity with graph neural networks, Bioinformatics. 37 (8) (2021) 1140–1147.
[24] Y. Zhang, J. Li, S. Lin, J. Zhao, Y. Xiong, D.Q. Wei, An end-to-end method for predicting compound-protein interactions based on simplified homogeneous graph convolutional network and pre-trained language model, J. Cheminf. 16 (1) (2024) 67.
[25] Z. Zhou, J. Chen, S. Lin, L. Hong, D.Q. Wei, Y. Xiong, GRATCR: epitope-specific T cell receptor sequence generation with data-efficient pre-trained models, IEEE J. Biomed. Health Inform. (2025).
[26] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, Adv. Neural Inf. Process. Syst. 30 (2017).
[27] L. Chen, X. Tan, D. Wang, F. Zhong, X. Liu, T. Yang, X. Luo, K. Chen, H. Jiang, M. Zheng, TransformerCPI: improving compound–protein interaction prediction by sequence-based deep learning with self-attention mechanism and label reversal experiments, Bioinformatics 36 (16) (2020) 4406–4414.
[28] L. Huang, J. Lin, R. Liu, Z. Zheng, L. Meng, X. Chen, X. Li, K.C. Wong, CoaDTI: Multi-modal co-attention-based framework for drug–target interaction annotation, Briefings Bioinf. 23 (6) (2022) bbac446.
[29] H. Kurata, S. Tsukiyama, ICAN: interpretable cross-attention network for identifying drug and target protein interactions, PLoS One 17 (11) (2022) e0276609.
[30] M. Wang, J. Wang, Z. Rong, L. Wang, Z. Xu, L. Zhang, K. Li, A bidirectional interpretable compound–protein interaction prediction framework based on cross attention, Comput. Biol. Med. 172 (2024) 108239.
[31] Z. Hou, Z. Xu, C. Yan, H. Luo, J. Luo, CPI-GGS: a deep learning model for predicting compound–protein interaction based on graphs and sequences, Comput. Biol. Chem. 115 (2025) 108326.
[32] A. Maroua, G. Tian, R. Wang, J. Zhou, TriCvT-DTI: predicting drug–target interactions using trimodal representations and convolutional vision transformers, IEEE J. Biomed. Health Inform. (2025).
[33] Y. Fang, Y. Jiang, L. Wei, Q. Ma, Z. Ren, Q. Yuan, D.Q. Wei, DeepProSite: structure-aware protein binding site prediction using ESMFold and pretrained language model, Bioinformatics. 39 (12) (2023) btad718.
[34] X. Mao, Y. Chu, D. Wei, Designed with interactome-based deep learning, Nat. Chem. Biol. 20 (11) (2024) 1399–1401.
[35] P. Veličković, A. Casanova, P. Liò, G. Cucurull, A. Romero, Y. Bengio, Graph attention networks, in: 6th International Conference on Learning Representations (ICLR 2018) – Conference Track Proceedings, 2018 arXiv:1710.10903.
[36] P. Li, Y. Li, C.Y. Hsieh, S. Zhang, X. Liu, H. Liu, S. Song, X. Yao, A comparative study of deep learning-based drug–target interaction prediction methods, Briefings Bioinf. 22 (4) (2021) bbaa266.
[37] S. Li, F. Wan, H. Shu, T. Jiang, J. Zeng, MONN: a multi-objective neural network for predicting compound–protein interactions and affinities, Cell Syst. 10 (4) (2020) 308–322.e11.
[38] H. Liu, J. Sun, J. Guan, J. Zheng, S. Zhou, Improving compound–protein interaction prediction by building up highly credible negative samples, Bioinformatics 31 (12) (2015) i221–i229.
[39] F. Zhu, Z. Shi, C. Qin, L. Tao, X. Liu, F. Xu, L. Zhang, Y. Song, X. Liu, J. Zhang, B. Han, P. Zhang, Y. Chen, Therapeutic target database update 2012: a resource for facilitating target-oriented drug discovery, Nucleic Acids Res. 40 (D1) (2012) D1128–D1136.
[40] A.P. Bento, A. Hersey, E. Félix, G. Landrum, A. Gaulton, F. Atkinson, L.J. Bellis, M. De Veij, A.R. Leach, An open source chemical structure curation pipeline using RDKit, J. Cheminf. 12 (2020) 51.
[41] Z. Yang, W. Zhong, L. Zhao, C.Y.C. Chen, MGraphDTA: deep multiscale graph neural network for explainable drug–target binding affinity prediction, Chem. Sci. 13 (3) (2022) 816–833.
[42] Y. Li, P. Li, X. Yang, C.Y. Hsieh, S. Zhang, X. Wang, R. Lu, H. Liu, X. Yao, Geometric deep learning of molecular structures for drug discovery, Chem. Eng. J. 414 (2021) 128817.
[43] A. Rasool, J.U. Rahman, Q. Iqbal, Optimizing GNN architectures through nonlinear activation functions for potent molecular property prediction, Computation 12 (11) (2024) 212.
[44] K. Abbasi, P. Razzaghi, A. Poso, M. Amanlou, J.B. Ghasemi, A. Masoudi-Nejad, DeepCDA: deep cross-domain compound–protein affinity prediction through LSTM and convolutional neural networks, Bioinformatics 36 (17) (2020) 4633–4642.
[45] B. Małysiak-Mrozek, T. Baron, D. Mrozek, Spark-IDPP: high-throughput and scalable prediction of intrinsically disordered protein regions with Spark clusters on the cloud, Clust. Comput. 22 (2) (2019) 487–508.
[46] B. Małysiak-Mrozek, D. Mrozek, An improved method for protein similarity searching by alignment of fuzzy energy signatures, Int. J. Comput. Intell. Syst. 4 (1) (2011) 75–88.
[47] R. Kumar, A. Indrayan, Receiver operating characteristic (ROC) curve for medical researchers, Indian Pediatr. 48 (4) (2011) 277–287.
[48] B. Li, J. Tang, Q. Yang, S. Li, X. Cui, Y. Li, Y. Chen, W. Xue, X. Li, F. Zhu, NOREVA: normalization and evaluation of MS-based metabolomics data, Nucleic Acids Res. 45 (W1) (2017) W162–W170.
[49] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, H. Adam, Encoder-decoder with atrous separable convolution for semantic image segmentation, in: V. Ferrari, M. Hebert, C. Sminchisescu, Y. Weiss (Eds.), Proceedings of the European Conference on Computer Vision (ECCV), Springer, 2018, pp. 833–851.
[50] P. Li, Y. Li, C.Y. Hsieh, S. Zhang, X. Liu, H. Liu, X. Yao, TrimNet: learning molecular representation from triplet messages for biomedicine, Briefings Bioinf. 22 (4) (2021) bbaa280.
[51] L. Wei, W. Long, L. Wei, MDL-CPI: multi-view deep learning model for compound–protein interaction prediction, Methods 204 (2022) 418–427.
[52] X. Lin, Z. Quan, Z. Wang, Y. Guo, X. Zeng, P.S. Yu, Effectively identifying compound–protein interaction using graph neural representation, IEEE ACM Trans. Comput. Biol. Bioinf 20 (2) (2023) 1157–1168.